

# Logic Circuits Unveiled: Bridging the Gap between Discrete Mathematics and Computer Science Education

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# • Who are we? •



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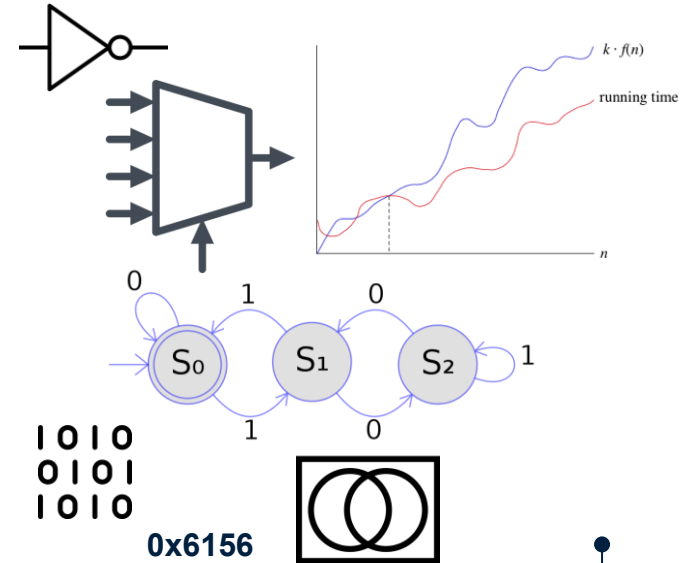
# CPSC 121

- First-year introductory discrete mathematics course for Computer Science students. Covers fundamental theory such as:

- Mathematical Logic
- Number Representation
- Finite State Machines + Regular Expressions
- Formal Proofs (incl. Induction)
- Big O
- Computer Architecture

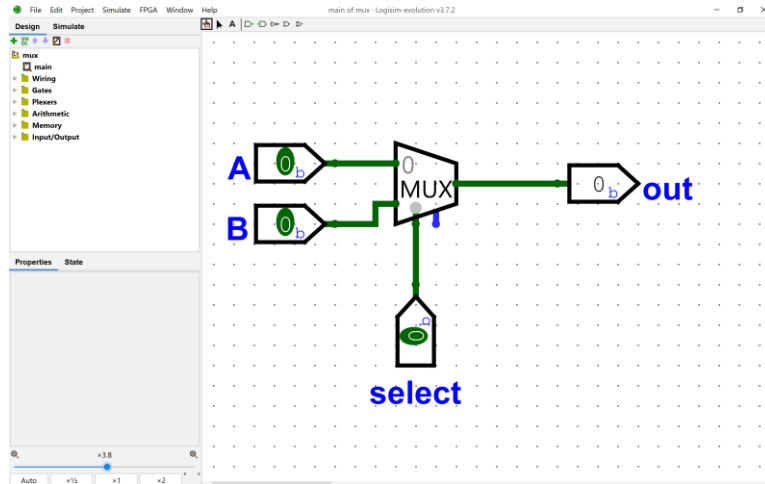
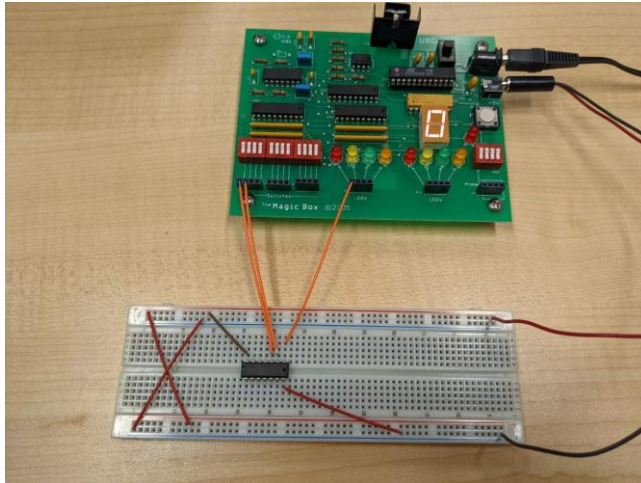
- Large active-learning approach:

- Pre-reading quizzes
- Clickers & Worksheets in-lecture
- Tutorials (Guided problem-solving)
- Digital Logic-focused Labs



# CPSC 121 Labs

- Labs focus on digital logic applications of material from lecture using a simple hardware kit called a **Magic Box** and a digital logic simulator called **Logisim**.
- Consisted of a pre-lab assignment + 2-hour lab session.



# Existing Issues

- Cohesiveness
  - Within-lab cohesiveness
  - Lab-lecture cohesiveness
- Labs specifically:
  - Equipment Issues (Software & Hardware)
  - Lack of supporting resources outside of labs
- Changes were made to address these issues, implemented over time – starting in 2022 and finalized in 2023.

# Resources - Lab Notes & Screencasts

- Lab Notes were developed to support students:
  - More traditional study materials
  - Provided learning objectives, review of material, practice problems
- Screencasts were developed for pre-labs:
  - Short, traditional lecture format focusing on main topics for each lab
- Overall, this allowed us to better support students and emphasize the importance of this content



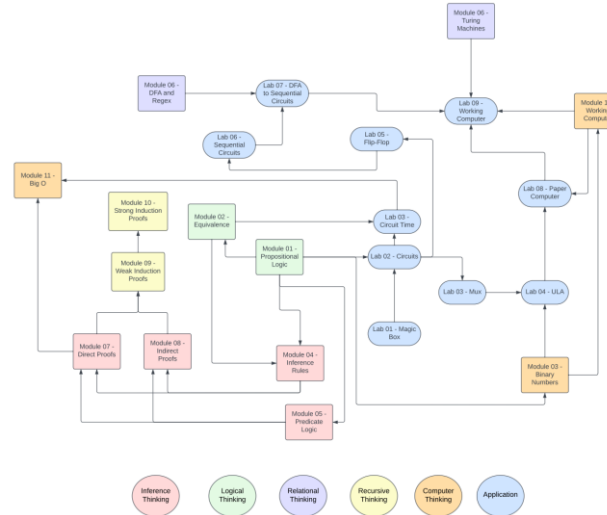
# Reworking Lab Content

- Content was reworked to be more cohesive within labs and between labs and lecture.
- Example: Lab 4
  - Previously provided disjointed exercises involving MUXs (already discussed in Labs 2&3) and an ALU
  - We replaced the less relevant tasks with content about binary numbers and data representation, including adders and RAM
  - This tied better with lecture content recently covered, and led into the next lab better which focused on memory



# Concept Map

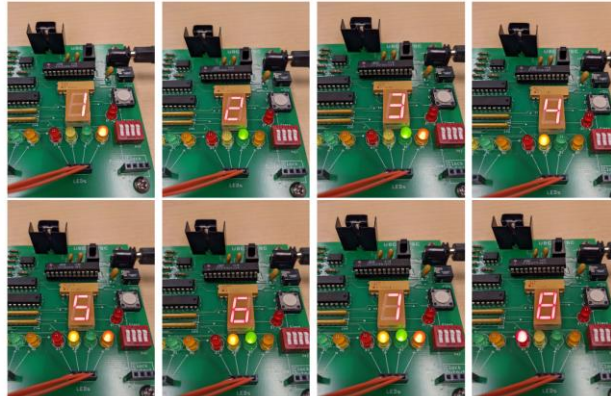
- A concept map was shown to students at the start of every new module (weekly) to center this content relative to other units.
- Units were color-coded to focus on high-level skills developed, focusing on the “bigger picture”



# Lecture Connection

- Finally, whenever content in lecture tied well to labs, we made a point to connect it for students.

7-Segment Display

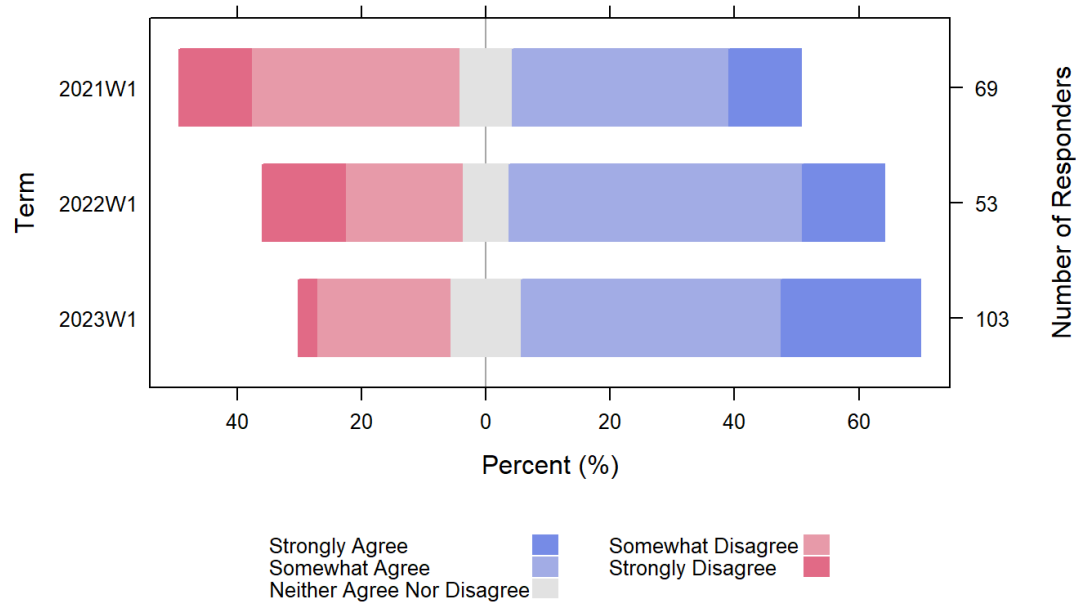


# Summary of Changes

- Changes made:
  - Added additional lab resources (Lab Notes + Screencasts)
  - Updated lab content (e.g., Lab 4)
  - Concept map
  - Actively connected lecture to lab whenever possible
  - Updated software from Logisim to Logisim Evolution, a newer version
- We wanted to evaluate how well these improved cohesiveness and effectiveness of the course. Data was collected through end-of-lab surveys and end-of-term surveys.

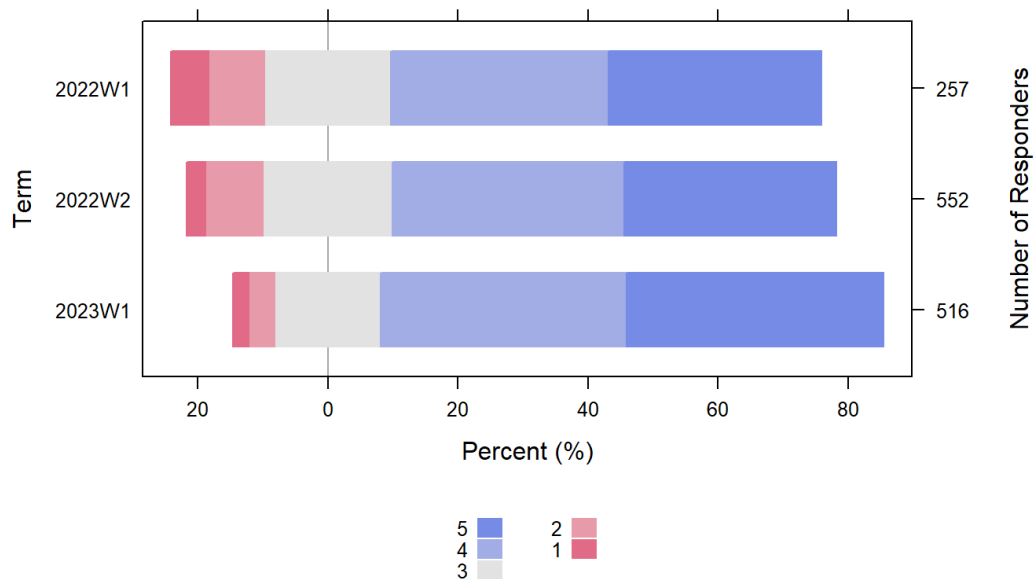
# Results – Lab-Lecture Cohesiveness

"I feel that lecture and lab content is well integrated."



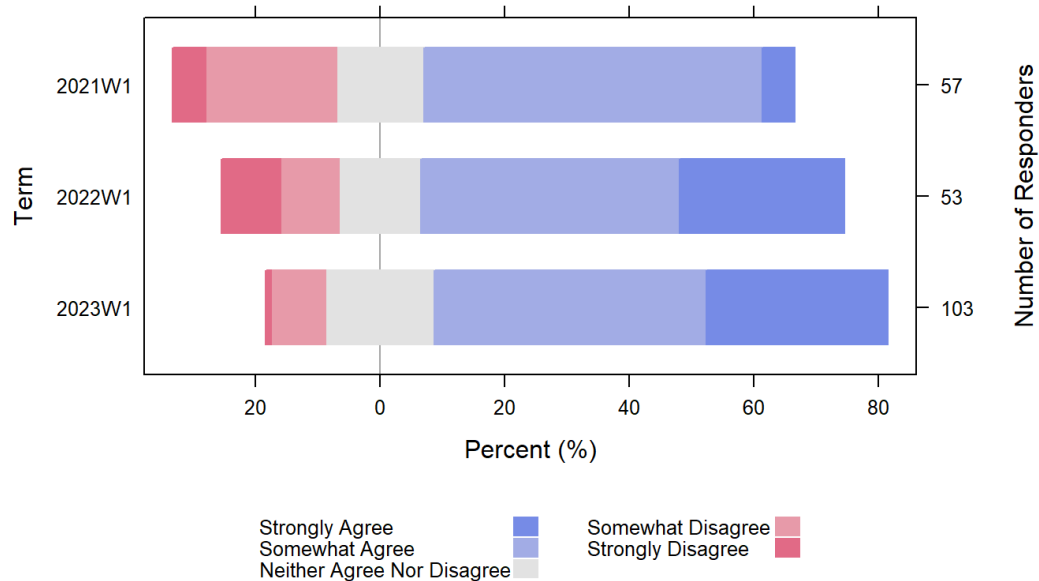
# Results – Within-Lab Cohesiveness

"The lab content throughout this course felt cohesive."



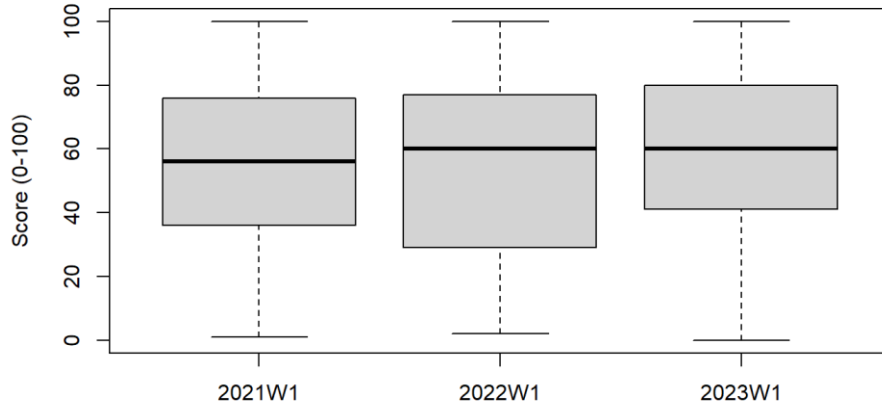
# Results – Lab-Course Cohesiveness

"Overall, I think labs were helpful to my learning of the course."

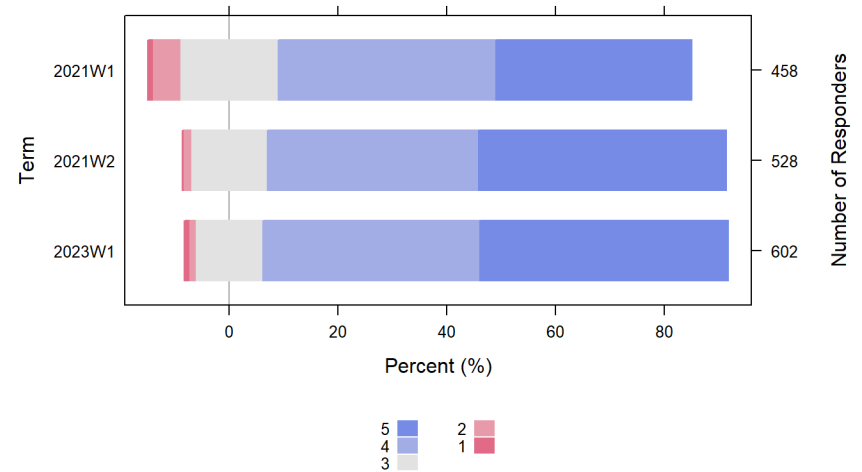


# Results – Lab Effectiveness

How effective was each part of the course for your learning... Labs including pre-labs



"The lab was an effective learning exercise. (Lab 4)"





# Results – Lab Effectiveness cont.

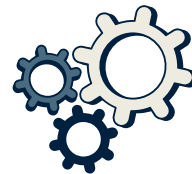
- Effectiveness ratings seemed related to enjoyment of labs.
  - Students who rated labs poorly often mentioned equipment issues in open-response sections of surveys
  - Students who rated labs highly effective often mentioned having great TAs that contributed to their lab experience.
- Improving effectiveness of labs further may therefore require focusing on these factors instead, to allow students to fully focus on learning rather than external issues.

# Results – Active Learning & Grading

- Qualitative feedback highlighted student concerns related to active learning used in labs:
  - Students commented on labs generally being harder content than lecture as less time was dedicated to them. Frequency of these comments decreased after traditional resources like lab notes were introduced.
  - Students often asked to address lab content in a lecture after lab.
  - Directing students to office hours may address these issues.
- Students also commented that labs were highly stressful, due to the time limit.
  - This is likely as the grading structure may encourage students to focus on grades over learning, even with grade averages high in labs (90%+ averages)
  - Revisiting the grading scheme for labs + the course may address this

# Conclusion

- Our changes to improve cohesiveness had direct impacts on how students viewed lab content and engaged with it.
- It's necessary to actively connect concepts for first-year students who don't have the perspective of where discrete math content may be relevant or applicable.



# Next Steps



## Upgrade Magic Boxes

- Use a more reliable platform that may lead to less equipment issues for students.
- This may also open up opportunities to tie content into other modules (e.g., proofs)



## Revising of Grading Scheme

- Revising lab/course grading scheme to reduce stress may allow them to focus more on learning.
- E.g., removing “pass lab” component



## Auto-generation & Randomization of questions

- Allows us to address concerns about “not enough practice” and open up more innovative approaches to grading.



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